



Tests for proportionality of matrices with large dimension

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ABSTRACT

A test for proportionality of two covariance matrices with large dimension, possibly larger than the sample size, is proposed. The test statistic is simple, computationally efficient, and can be used for a large class of multivariate distributions including normality. The properties of the statistic, including asymptotic distribution, are given under high-dimensional set up. Through simulations, the statistic is shown to perform accurately, and outperform its recent competitors, constructed on the basis of similar principles. An extension to the multi-sample case is given.

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1. Introduction

Let $\mathbf{x}_{ik} \in \mathbb{R}^p$, $k \in \{1, \dots, n_i\}$, be two independent random samples of n_i iid vectors in i th sample, $i \in \{1, 2\}$, generated by a probability space $(\mathcal{X}, \mathcal{A}, \mathcal{P})$ where $\mathcal{X}$ is the sample space, $\mathcal{A}$ is the σ -field and $\mathcal{P}$ denotes an appropriate probability measure. We let $\mathcal{P}$ refer to a Lebesgue density $\mathcal{P}_{\theta_i}$, where $\theta_i \in \{\mu_i, \Sigma_i\}$ denotes the parameter space composed of mean vector, $E(\mathbf{x}_{ik}) = \mu_i \in \mathbb{R}^p$, and covariance matrix, $\Sigma_i = E(\mathbf{x}_{ik} - \mu_i)(\mathbf{x}_{ik} - \mu_i)^T \in \mathbb{R}^{p \times p}$, assumed positive-definite.

Tests of elements of parameter space, particularly covariance operators, Σ_i , are of frequent interest in multivariate theory. An important motivation for such tests is that they help reduce the dimension of the underlying space, simplifying the mathematical and computational aspects of the analysis. This motivation, hence the corresponding testing theory, takes special importance for high-dimensional data, in particular when $p \gg n_i$. For example, a test of sphericity, $\Sigma_i \propto \mathbf{I}$, where $\mathbf{I}$ denotes an identity matrix, helps drastically reduce the number of parameters to be estimated.

For two or more samples, similar objectives are achieved by testing hypotheses such as homogeneity or proportionality of several covariance matrices. Our objective in this paper is to address the latter case, i.e., of proportionality of covariance matrices. We are interested to construct a test statistic for

$$H_0 : \Sigma_1 = \delta \Sigma_2 \text{ vs. } H_1 : \Sigma_1 \neq \delta \Sigma_2 \quad (1)$$

when the data are high-dimensional ($p \gg n_i$) and $\mathcal{P}_{\theta_i}$ may not necessarily be multivariate normal. Here, $\delta > 0$ denotes the constant of proportionality. It reduces the testing problem in (1) to that of homogeneity test when $\delta = 1$. The proportionality hypothesis plays a central role in several multivariate techniques, e.g., in discriminant and cluster analyses, and is also considered as a close alternative when the homogeneity assumption is not a viable option.

Test of hypothesis in (1) for the classical case, i.e., for $p < n$, $n = n_1 + n_2$, is mostly based on the likelihood-ratio testing framework. Probably the oldest work in this direction is [11], later extended by [10]. [13] discussed both estimation and testing for multi-sample case, where [26] gives a Wald type test for H_0 . However, the requirement of the use of

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determinants or inverses of empirical covariance matrices renders the classical inference inapplicable for high-dimensional data whence such empirical estimators become singular. This calls for new realms of testing theory to be considered.

This new realm, for testing large covariance matrices, is mostly based on a Frobenius norm, measuring the discrepancy between the null and alternative hypothesis, e.g., $\|\Sigma_1 - \Sigma_2\|^2$, for homogeneity tests ($\delta = 1$ in H_0). But, unlike homogeneity test, where the norm reduces to a simple linear combination of trace operators, the situation becomes a bit tricky for proportionality testing because of the constant δ involved. For a detailed study of this problem, see e.g., [17,23], or a brief discussion in [13].

Still, a few attempts have recently been made on testing problem (1) for high-dimensional data; e.g., [8,25], where the latter connect the problem in (1) to testing spatial sign covariance matrices. Of particular mention for us is a recent work by [29] who address (1) using a simple but elegant idea which helps get rid of the proportionality constant δ , resulting into an equivalent homogeneity testing problem, for a slightly re-scaled matrices. Apart from its use in testing problem (1), their idea has further intuitive ramifications, as highlighted in the subsequent section. More recently, [9] introduced a statistic, using a similar idea of re-scaled matrices to avoid the proportionality constant.

We, in our attempt and appreciation of [29] approach, use a simple result from linear algebra, which likewise helps get rid of δ , and propose a new test for (1). The test is much simpler than that of [29], or of [9], is more intuitive and computationally much less expensive, has similar limit distribution, does not need normality assumption, and, as shown through simulations, performs reasonably better.

After a short introduction to the aforementioned tests in [9,29], the new test and its properties are discussed in Section 2, with a multi-sample extension in Section 3. An evaluation of the proposed test using extensive simulations, including a detailed comparison with [9,29] tests, is given in Section 4. A summary in Section 5 concludes the paper, where technical details are collected in the appendix.

Before proceeding with this outline, it may be mentioned that the testing strategy presented here can be readily extended to functional data. In fact, it mainly needs extending the language and notation from matrices to operators. To see this, recall the probability space $(\mathcal{X}, \mathcal{A}, \mathcal{P})$ from Section 1, and embed $\mathcal{P}$ into an $\mathcal{L}_2(\cdot)$ space, $\mathcal{L}_2(\mathcal{P})$, call it $\mathcal{H}$, which is a space over $\mathbb{R}$, equipped with an inner product $\langle \mathbf{x}, \mathbf{y} \rangle : \mathcal{H} \times \mathcal{H} \rightarrow \mathbb{R}$, so that $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$ is its (squared) norm. Assuming $E\|\mathbf{x}\|^2 < \infty$, $\mathcal{H}$ is a Hilbert space, hence has an orthonormal basis. This is functional data set up and the theory presented here can be accommodated to construct a test, including for (1).

Tests of such hypotheses, particularly for equality of covariance operators, have recently been considered. A case of two samples, focusing on asymptotic framework, is given in [14,15], with a multi-sample extension in [6]. Test statistics on covariance operators for functional data usually replace the Frobenius norm as a criterion to measure the discrepancy between the null and alternative hypothesis with Hilbert–Schmidt norm. Further, like in classical multivariate analysis, such tests are often also needed to determine the nature of other tests of interest, e.g., tests on mean functions. [21] discuss such a test when data are defined on an $\mathcal{L}_2([0, 1])$ space; for more details, see [20], who also use the theory in the context of spatial inference.

As functional data analysis can comfortably handle the case of finite and infinite dimensional spaces, the inferential theory developed within its realm finds extensive modern applications; for a typical case of functional MRI data, particularly from the perspective of many testing problems, see [7]. For more details, see [5], where special reviews in this direction have recently been put forth in [12,16].

2. Test statistic and its properties

For $\mathbf{x}_{ik} \in \mathbb{R}^p$, defined above, let the data matrix $\mathbf{X}_i = (\mathbf{x}_{i1}^\top, \dots, \mathbf{x}_{in_i}^\top)^\top \in \mathbb{R}^{n_i \times p}$ collect all n_i vectors, with $E(\mathbf{x}_i) = \mathbf{1}_{n_i} \otimes \boldsymbol{\mu}_i$, $\text{Cov}(\mathbf{x}_i) = \mathbf{I}_{n_i} \otimes \Sigma_i$, where $\otimes$ denotes the Kronecker product. The unbiased estimators of $\boldsymbol{\mu}_i$ and Σ_i , i.e., $\bar{\mathbf{x}}_i = \sum_{k=1}^{n_i} \mathbf{x}_{ik}/n_i$ and $\hat{\Sigma}_i = \sum_{k=1}^{n_i} (\tilde{\mathbf{x}}_{ik} \tilde{\mathbf{x}}_{ik}^\top)/(n_i - 1)$, $\tilde{\mathbf{x}}_{ik} = \mathbf{x}_{ik} - \bar{\mathbf{x}}_i$, can be compactly represented, using $\mathbf{X}_i$ matrices, $i \in \{1, 2\}$, as

$$\bar{\mathbf{x}}_i = \frac{1}{n_i} \mathbf{X}_i^\top \mathbf{1}_{n_i}, \quad \hat{\Sigma}_i = \frac{1}{n_i - 1} \mathbf{X}_i^\top \mathbf{C}_{n_i} \mathbf{X}_i, \quad (2)$$

where $\mathbf{C}_{n_i} = \mathbf{I}_{n_i} - \mathbf{J}_{n_i}/n_i$ is the centering matrix with $\mathbf{I}$ as identity matrix of order n_i and $\mathbf{J}_{n_i} = \mathbf{1}_{n_i} \mathbf{1}_{n_i}^\top$ with $\mathbf{1}_{n_i}$ an $n_i \times 1$ vector of 1s. In the following, $\|\mathbf{a}\|^2$ and $\|\mathbf{A}\|^2 = \mathbf{A}^\top \mathbf{A}$ denote, respectively, the squared norm of a vector $\mathbf{a} \in \mathbb{R}^p$ and squared Frobenius norm of a real matrix $\mathbf{A} \in \mathbb{R}^{p \times q}$.

To construct a test statistic for (1), [29] use the following argument. As $\Sigma_1 = \delta \Sigma_2 \Rightarrow \text{tr}(\Sigma_1) = \delta \text{tr}(\Sigma_2) \Rightarrow \delta = \text{tr}(\Sigma_1)/\text{tr}(\Sigma_2)$, the testing problem can be equivalently stated as $\Sigma_1/\text{tr}(\Sigma_1) = \Sigma_2/\text{tr}(\Sigma_2)$. Denoting $\Sigma_i^{1/2} = \Delta_i$, the equivalent form implies testing homogeneity of normed covariance matrices, i.e., $H_0 : \Gamma_1 = \Gamma_2$, where $\Gamma_i = \Delta_i^2/\|\Delta_i\|^2 = \Sigma_i/\text{tr}(\Sigma_i)$, $i \in \{1, 2\}$. In fact, in this context, a general result can be stated as follows: Given two matrices $\mathbf{A}$ and $\mathbf{B}$ of the same order such that $\mathbf{A} \propto \mathbf{B}$. Then $\mathbf{A} = \mathbf{B}$ if and only if $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{B})$.

Using the aforementioned Frobenius norm approach for the equivalent form of (1), [29] consider $\|\Gamma_1 - \Gamma_2\|^2$ which, after re-scaling the components by p , is finally formulated as

$$\tau_{\text{TM}} = \sum_{i=1}^2 \epsilon_i^{-1} - 2\epsilon_{12}^{-1}, \quad (3)$$

where $\epsilon_i^{-1} = a_{i,2}/a_{i,1}^2$, $\epsilon_{12}^{-1} = a_{12}/a_{1,1}a_{2,1}$ with $a_{i,1} = \text{tr}(\Sigma_i)/p$, $a_{i,2} = \text{tr}(\Sigma_i^2)/p$ and $a_{12} = \text{tr}(\Sigma_1\Sigma_2)/p$. The plug-in estimators of the traces are then used to define the test statistic for (1) as

$$T_{\text{TM}} = \sum_{i=1}^2 \widehat{\epsilon}_i^{-1} - 2\widehat{\epsilon}_{12}^{-1}, \quad (4)$$

with $\widehat{\epsilon}_i^{-1} = \widehat{a}_{i,2}/\widehat{a}_{i,1}^2$ and $\widehat{\epsilon}_{12}^{-1} = \widehat{a}_{12}/\widehat{a}_{1,1}\widehat{a}_{2,1}$, where $\widehat{a}_{i,1}$ and $\widehat{a}_{12}$ are unbiased and consistent estimators of $a_{i,1}$ and a_{12} , defined, along with $\widehat{a}_{i,2}$, as

$$\widehat{a}_{i,1} = \frac{1}{p} \text{tr}(\widehat{\Sigma}_i), \quad \widehat{a}_{i,2} = \frac{(n_i - 1)^2}{p(n_i - 2)(n_i + 1)} \left\{ \text{tr}(\widehat{\Sigma}_i^2) - \frac{[\text{tr}(\widehat{\Sigma}_i)]^2}{n_i - 1} \right\}, \quad \widehat{a}_{12} = \frac{1}{p} \text{tr}(\widehat{\Sigma}_1 \widehat{\Sigma}_2).$$

This formulation provides another intuition pertaining to (1). Recall the equivalent form $\Sigma_1/\text{tr}(\Sigma_1) = \Sigma_2/\text{tr}(\Sigma_2)$. Pre-multiplying by Σ_1 , post-multiplying by Σ_2 , and simplifying, we can write (1) in terms of ϵ 's as $\epsilon_1^{-1} = \epsilon_2^{-1} \Leftrightarrow \epsilon_1 = \epsilon_2$.

Further, for any positive semi-definite matrix of order, say p , $\epsilon = [\text{tr}(\mathbf{A})]^2/p \text{tr}(\mathbf{A}^2)$ is a well-known measure of sphericity, so that $\mathbf{A} \propto \mathbf{I} \Leftrightarrow \epsilon = 1$, where extreme non-sphericity, e.g., $\mathbf{A}_p = \mathbf{1}\mathbf{1}^\top$, gives $\epsilon = 1/p$ with $\mathbf{I}$ as identity matrix of order p and $\mathbf{1}$ a $p \times 1$ vector of 1s. In τ_{TM} , $\epsilon_i^{-1} = p\|\Gamma_i\|^2$ with $\epsilon_i = [\text{tr}(\Sigma_i)]^2/p \text{tr}(\Sigma_i^2)$. Likewise, ϵ_{12} can be considered as a similar two-sample measure, where $\epsilon_{12}^{-1} = p\|\Phi_1\Phi_2\|^2$, $\Phi_i = \Gamma_i^{1/2}$. Under H_0 , ϵ_{12} reduces to ϵ_i , making $\tau_{\text{TM}} = 0$. [32] give some interesting results on ratios like ϵ_i . Under normality, [29] show that, as $p \rightarrow \infty$ with n_i in certain orders of p ,

$$\frac{n_1 n_2}{n_1 + n_2} \frac{T_{\text{TM}}}{\widehat{b}_2} \xrightarrow{\mathcal{D}} N(0, 1), \quad (5)$$

under H_0 , where $\widehat{b}_2 = \sum_{i=1}^2 w_i \widehat{\epsilon}_i^{-1}$, $w_i = n_i^2/(n_1^2 + n_2^2)$, $i \in \{1, 2\}$. The performance of this limit is shown through simulations under several parameter settings; see Section 4 for details and comparison. The statistic T_{TM} is nice and seems to perform accurately, under high-dimensional setting. In particular, the formulation of the statistic does not depend on the proportionality constant δ .

Following a similar approach, as in [29], [9] also introduce a statistic for (1). They use plug-in estimators for the criterion

$$\tau_{\text{CLTZ}} = p^2 \|\Gamma_1 - \Gamma_2\|^2$$

and define their test statistic as

$$T_{\text{CLTZ}} = p^2 \|\widetilde{\Gamma}_1 - \widetilde{\Gamma}_2\|^2, \quad (6)$$

with $\widetilde{\Gamma}_i = \widehat{\Sigma}_i/\text{tr}(\widehat{\Sigma}_i)$ denoting a plug-in estimator of $\Gamma_i = \Sigma_i/\text{tr}(\Sigma_i)$, $i \in \{1, 2\}$. Although a close similarity between τ_{CLTZ} and τ_{TM} , correspondingly between T_{CLTZ} and T_{TM} , particularly in the limit, is obvious, the moments of T_{CLTZ} and their estimators, as given in [9], are much more involved than those for T_{TM} . Under certain conditions, particularly involving up to fourth order moments of $\mathbf{x}_{ik}$ and relationship between n_i and p , [9] show the limit of their statistic as

$$(T_{\text{CLTZ}} - E(T_{\text{CLTZ}}))/\sqrt{\text{Var}(T_{\text{CLTZ}})} \xrightarrow{\mathcal{D}} N(0, 1)$$

with mean and variance under $H_0 : \Sigma_1 = \delta \Sigma_2 = \Sigma$ given, respectively, as

$$\mu_0 = p \sum_{i=1}^2 \frac{1}{n_i(n_i - 1)^2} \{n_i(n_i + 1)\epsilon^{-1} + n_i^2 \beta_{z_i} \kappa + p(n_i^2 - n_i - 1)\}, \quad \sigma_0^2 = 4p^2 \left(\frac{n}{n_1 n_2} \right) \epsilon^{-2},$$

where ϵ^{-1} is same as ϵ_i for τ_{TM} in (3) but defined for Σ under H_0 , $\kappa = \sum_{j=1}^p \sigma_j^4 / (\sum_{j=1}^p \sigma_j^2)^2$ with σ_j^2 the diagonal elements of Σ , β_{z_i} correspond to fourth moments of $\mathbf{z}_{ik}$ in (7) [see Condition 1 in 9], and $n = \sum_{i=1}^2 n_i$. The moments are estimated as

$$\begin{aligned} \widehat{\mu}_0 &= p \sum_{i=1}^2 \frac{1}{n_i(n_i - 1)^2} \left\{ n_i(n_i + 1) \left(\widetilde{\epsilon}_i^{-1} - \frac{p}{n_i} \right) + n_i^2 \widehat{\kappa}_i + p(n_i^2 - n_i - 1) \right\}, \\ \widehat{\sigma}_0^2 &= 4p^2 \left(\frac{n}{n_1 n_2} \right) \sum_{i=1}^2 \frac{1}{n_i} \left(\widetilde{\epsilon}_i^{-1} - \frac{p}{n_i} \right)^2, \end{aligned}$$

where $\widetilde{\epsilon}_i = p\|\widetilde{\Gamma}_i\|^2$, and we denote $\widehat{\kappa}_i = \widehat{a}_i/(p^{-1} \text{tr}(\widehat{\Sigma}_i))^2$ with $\widehat{a}_i$, $i \in \{1, 2\}$, having long expressions; see Section 2 and Appendix in [9] for $\widehat{a}_i$ and other detailed computations.

In the following, we use a simple result from linear algebra, which also helps dispense with δ , and use it to construct a statistic for the testing problem (1). The statistic is very simple and is composed of computationally very efficient estimators of the traces involved. Further, we also relax normality assumption and show that the proposed statistic can be used for a wider class of multivariate models. Through extensive simulations, we show an accurate performance of the proposed test, where comparisons with the aforementioned two statistics are also given.

To proceed with this, we begin by defining the general model which helps us relax normality. Given the data vectors $\mathbf{x}_{ik}$, $k \in \{1, \dots, n_i\}$, let $\mathbf{y}_{ik} = \mathbf{x}_{ik} - \boldsymbol{\mu}_i$, $i \in \{1, 2\}$, and define

$$\mathbf{y}_{ik} = \boldsymbol{\Sigma}_i^{1/2} \mathbf{z}_{ik}, \quad (7)$$

with $\mathbf{z}_{ik} \sim \mathcal{F}_i$, $E(\mathbf{z}_{ik}) = \mathbf{0}_p$, $\text{Cov}(\mathbf{z}_{ik}) = \mathbf{I}_p \forall i$, where $\mathcal{F}$ denotes a p -dimensional distribution function, $\mathbf{0}_p$ is a vector of zeros and $\mathbf{I}_p$ is the identity matrix. Model (7) is quite general and flexible, includes multivariate normality as a special case, and will help us study the properties of the test statistic under a few mild assumptions.

The construction of this test statistic is based on the following simple lemma, which is nothing but the Cauchy–Schwarz inequality for matrix traces; see e.g., [27, p. 262].

Lemma 1. Let $\mathbf{A}_i \in \mathbb{R}^{p \times q}$, $i \in \{1, 2\}$, be two real matrices. Then

$$[\text{tr}(\mathbf{A}_1^\top \mathbf{A}_2)]^2 \leq \text{tr}(\mathbf{A}_1^\top \mathbf{A}_1) \text{tr}(\mathbf{A}_2^\top \mathbf{A}_2),$$

where equality holds if and only if $\mathbf{A}_1 = \delta \mathbf{A}_2$ with $\delta > 0$ a constant.

The necessary and sufficient condition for equality in Lemma 1 helps us avoid δ in (1) and write it equivalently as

$$\boldsymbol{\Sigma}_1 = \delta \boldsymbol{\Sigma}_2 \Leftrightarrow \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2) / \sqrt{\text{tr}(\boldsymbol{\Sigma}_1^2) \text{tr}(\boldsymbol{\Sigma}_2^2)} = 1.$$

The last expression can also be written as $\text{tr}(\tilde{\boldsymbol{\Sigma}}_1 \tilde{\boldsymbol{\Sigma}}_2) = 1$ where $\tilde{\boldsymbol{\Sigma}}_i = \boldsymbol{\Sigma}_i / \|\boldsymbol{\Sigma}_i\| = \boldsymbol{\Delta}_i^2 / \|\boldsymbol{\Delta}_i^2\|$ is the normed version of $\boldsymbol{\Sigma}_i$, $i \in \{1, 2\}$. A test statistic for (1) can then be defined as

$$T = \text{tr}(\tilde{\boldsymbol{\Sigma}}_1 \tilde{\boldsymbol{\Sigma}}_2) - 1 = \frac{\|\widehat{\boldsymbol{\Delta}}_1 \widehat{\boldsymbol{\Delta}}_2\|^2}{\sqrt{\|\widehat{\boldsymbol{\Sigma}}_1\|^2 \|\widehat{\boldsymbol{\Sigma}}_2\|^2}} - 1. \quad (8)$$

The components of the statistic T are to be essentially considered with scaling by p^2 , i.e., $\|\widehat{\boldsymbol{\Delta}}_1 \widehat{\boldsymbol{\Delta}}_2\|^2 / p^2$ and $\|\widehat{\boldsymbol{\Sigma}}_i\|^2 / p^2$, so that, this scaling is an inevitable part of the components when we study the component separately. However, once plugged back in T , the effect of p^2 can be ignored.

The statistic T has a very simple form and, unlike T_{TM} , T is defined for any arbitrary δ in H_0 . Further, as shown below, T is essentially based on $\|\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2\|^2 = \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)$ only, using consistent estimators for the denominator; i.e., T defines itself by the numerator, where a consistent estimator of the denominator ensures its stable limit under high-dimensional setting. We begin by defining these estimators, where we denote $\|\widehat{\boldsymbol{\Delta}}_1 \widehat{\boldsymbol{\Delta}}_2\|^2 = E_{12}$ and $\|\widehat{\boldsymbol{\Sigma}}_i\|^2 = E_i$, $i \in \{1, 2\}$.

Definition 1. Assume Model (7) and let $\|\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2\|^2$ and $\|\boldsymbol{\Sigma}_i\|^2$, $i \in \{1, 2\}$, be as defined in the context of T in (8). The estimators of $\|\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2\|^2$ and $\|\boldsymbol{\Sigma}_i\|^2$ are defined, respectively, as

$$E_{12} = \|\widehat{\boldsymbol{\Delta}}_1 \widehat{\boldsymbol{\Delta}}_2\|^2, \quad (9)$$

$$E_i = v_i [(n_i - 1)(n_i - 2) \|\widehat{\boldsymbol{\Sigma}}_i\|^2 + [\|\widehat{\boldsymbol{\Delta}}_i\|^2]^2 - n Q_i], \quad (10)$$

where $v_i = (n_i - 1)/n_i(n_i - 2)(n_i - 3)$, $\widehat{\boldsymbol{\Sigma}}_i$ is given in (2) and $Q_i = \sum_{k=1}^{n_i} (\tilde{\mathbf{x}}_{ik}^\top \tilde{\mathbf{x}}_{ik})^2$ with $\tilde{\mathbf{x}}_{ik} = \mathbf{x}_{ik} - \bar{\mathbf{x}}_i$, $i \in \{1, 2\}$.

By the independence of samples, $\|\widehat{\boldsymbol{\Delta}}_1 \widehat{\boldsymbol{\Delta}}_2\|^2 = \text{tr}(\widehat{\boldsymbol{\Sigma}}_1 \widehat{\boldsymbol{\Sigma}}_2)$ as an unbiased estimator of $\|\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2\|^2 = \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)$ is obvious. The estimator $\|\widehat{\boldsymbol{\Sigma}}_i\|^2$ is likewise simple, composed only of empirical estimator. Several simple trace estimators of this form are defined in [19], further extended in [3]. The simplicity of these estimators extends to computational aspect, eventually also to the test statistic T in (8).

In B, an alternative formulation of the same estimators, using U -statistics with symmetric kernels, is given. Due to interesting projection and asymptotic properties of U -statistics, this formulation is more useful in studying theoretical properties of the estimators, as stated in Theorem 1. The proof of this theorem needs following assumptions under Model (7).

Assumption 1. For $\mathbf{y}_{ik} = (Y_{ik1}, \dots, Y_{ikp})^\top$, $E(Y_{iks}^4) = \gamma_{is} \leq \gamma < \infty$, $\forall s \in \{1, \dots, p\}$, $i \in \{1, 2\}$.

Assumption 2. For $p \rightarrow \infty$, $\text{tr}(\boldsymbol{\Sigma}_i)/p = O(1)$, $i \in \{1, 2\}$.

Assumption 3. For n_i , $p \rightarrow \infty$, $p/n_i \rightarrow \zeta_i \leq \zeta_0 \in (0, \infty)$, $i \in \{1, 2\}$.

Assumption 4. $\lim_{p \rightarrow \infty} \text{tr}(\boldsymbol{\Sigma}_i^a \odot \boldsymbol{\Sigma}_j^b) / \text{tr}(\boldsymbol{\Sigma}_i^a \otimes \boldsymbol{\Sigma}_j^b) = 0$, $a, b \in \{1, 2, 3\}$, $a + b \leq 4$, $i, j \in \{1, 2\}$, where $\odot$ and $\otimes$ are Hadamard and Kronecker products, i, j denote population index and a, b are exponents.

As a consequence of Assumption 2, $\|\boldsymbol{\Sigma}_i\|^2/p^2 = O(1)$, which will also help us deal with several components in the moments under Model (7). Assumption 1 is essential when we relax normality and work under the general model.

Assumption 2 holds for many practically useful covariance structures, including spiked cases like compound symmetry. **Assumption 3** is needed to control simultaneous ordering of n_i and p . **Assumption 4**, in conjunction with **Assumption 2** or its consequence, will help us let certain terms vanish and conveniently obtain the asymptotic distribution of T . Following theorem gives the properties of estimators.

Theorem 1. Under Model (7) and **Assumption 1**, the estimators in **Definition 1** are unbiased for their respective traces, with

$$\text{Var}(E_{12}) = \frac{2}{(n_1 - 1)(n_2 - 1)} \left[(n - 1) \text{tr}\{(\Sigma_1 \Sigma_2)^2\} + \{\text{tr}(\Sigma_1 \Sigma_2)\}^2 + M_2 O(n) + M_3 O(1) \right], \quad (11)$$

$$\text{Var}(E_i) = \frac{4}{P(n_i)} \left[a(n_i) \text{tr}(\Sigma_i^4) + b(n_i) \{\text{tr}(\Sigma_i^2)\}^2 + M_2 O(n_i^3) + M_3 O(n_i^2) \right], \quad (12)$$

$$\text{Cov}(E_{12}, E_i) = \frac{4}{n_i(n_i - 1)} \{n_i \text{tr}(\Sigma_i^3 \Sigma_j) + M_2 O(n_i)\}, \quad (13)$$

where $n = n_1 + n_2$, $a(n_i) = 2n_i^3 - 9n_i^2 + 9n_i - 16$, $b(n_i) = n_i^2 - 3n_i + 8$, $P(n_i) = n_i(n_i - 1)(n_i - 2)(n_i - 3)$, and M_2, M_3 are given in **Theorem 4**.

The terms involving M_2, M_3 etc. vanish exactly under normality, and asymptotically for Model (7) under the assumptions. Further,

$$\text{Var}(E_i / E(E_i)) \leq O(1/n_i),$$

which is uniformly bounded in terms of p , and vanishes for $n_i \rightarrow \infty$. Adding the consequence of **Assumption 2**, it also follows that $E_i / p^2 \xrightarrow{P} \|\Sigma_i\|^2 / p^2$, as $n_i, p \rightarrow \infty$. This consistency and the independence of two samples imply that the ratio

$$\frac{\|\widehat{\Delta}_1 \widehat{\Delta}_2\|^2}{\|\widehat{\Sigma}_1\| \|\widehat{\Sigma}_2\|} - 1 \quad (14)$$

has the same limit distribution as T in (8). Using this and the results of **Theorem 1**, it follows that the mean and variance of $\sqrt{n_1 n_2} T$, denoted μ and σ^2 , respectively, can be approximated, for $n_i, p \rightarrow \infty$, as

$$E(\sqrt{n_1 n_2} T) = \tau[1 + o(1)], \quad (15)$$

$$\text{Var}(\sqrt{n_1 n_2} T) = 4(\tau + 1)^2[1 + o(1)], \quad (16)$$

with $\tau = \|\Delta_1 \Delta_2\|^2 / \|\Sigma_1\| \|\Sigma_2\| - 1$. Note that, under H_0 , $\tau = 0$ whence (15)–(16) reduce, respectively, to 0 and 1. Following theorem summarizes the limit of T .

Theorem 2. For T given in (8), $\sqrt{n_1 n_2} T \xrightarrow{D} N(\mu, \sigma^2)$ under Model (7) and **Assumptions 1–4**, as $n_i, p \rightarrow \infty$, with μ, σ^2 given in (15)–(16). In particular, under H_0 , $\sqrt{n_1 n_2} T \xrightarrow{D} N(0, 1)$.

From the proof of **Theorem 2** in B, the essential component of the statistic, E_{12} , formulated as U -statistic, has a first-order degenerate kernel so that the limit follows from the second-order projections, which also provides the non-vanishing component of $\text{Var}(T)$. Note that, the numerator and denominator of τ can, respectively, be estimated by E_{12} and E_i , $i \in \{1, 2\}$. In Section 4, we also demonstrate that T performs very accurately for moderate samples and comparably very large dimension.

3. Multi-sample extension

An extension of the testing problem (1) for multi-sample case can be considered as

$$H_{0i} : \Sigma_i = \delta_i \Sigma_1 \text{ vs. } H_{1i} : \Sigma_i \neq \delta_i \Sigma_1 \quad (17)$$

for $i \in \{2, \dots, g\}$, where $\delta_i > 0$ are constants, and H_{1i} holds for at least one i . Extending the notation correspondingly, let $\mathbf{x}_{ik} \in \mathbb{R}^p$, $k \in \{1, \dots, n_i\}$, be iid vectors from i th population with distribution $\mathcal{F}_i$, $i \in \{1, \dots, g\}$. Let $\widehat{\mathbf{x}}_i, \widehat{\Sigma}_i$ in (2) be similarly defined for all g independent samples. The extension of T in (8) for multi-sample case, along with its properties and limit, is relatively uncomplicated, so that we restrict this section only to essential components. Following **Lemma 1**, the test statistic in (8) extends as

$$T_i = \frac{\|\widehat{\Delta}_1 \widehat{\Delta}_i\|^2}{\sqrt{\|\widehat{\Sigma}_1\|^2 \|\widehat{\Sigma}_i\|^2}} - 1, \quad (18)$$

for $i \in \{2, \dots, g\}$, so that we can consider a vector of $g - 1$ test statistics

$$\mathbf{T} = (T_2, \dots, T_g)^\top, \quad (19)$$

where the components $\widehat{\Gamma}_1 \widehat{\Gamma}_i = \text{tr}(\widehat{\Sigma}_1 \widehat{\Sigma}_i)$ and $\|\widehat{\Sigma}_i\|^2$ are defined similarly as for two-sample case. The properties of individual T_i also remain same, so that the main difference pertains to assessing the covariances $\text{Cov}(T_{1i}, T_{1i'})$, $i \neq i'$, $i, i' = \{2, \dots, g\}$. By the consistency of E_i/p^2 for $\|\Sigma_i\|^2/p^2$, we can again focus on the numerator terms, $E_{1i} = \text{tr}(\widehat{\Sigma}_1 \widehat{\Sigma}_i)$. Denoting $Q(n_i) = n_i(n_i - 1)$, we need

$$\text{Cov}(E_{1i}, E_{1i'}) = \frac{1}{16Q(n_1)Q(n_i)Q(n_1)Q(n_{i'})} \sum_{\substack{k=1 \\ k \neq r}}^{n_1} \sum_{\substack{r=1 \\ r \neq s}}^{n_1} \sum_{\substack{\ell=1 \\ \ell \neq s}}^{n_i} \sum_{\substack{s=1 \\ s \neq \ell}}^{n_i} \sum_{\substack{k=1 \\ k \neq r}}^{n_1} \sum_{\substack{r=1 \\ r \neq s}}^{n_1} \sum_{\substack{\ell=1 \\ \ell \neq s}}^{n_{i'}} \sum_{\substack{s=1 \\ s \neq \ell}}^{n_{i'}} \text{Cov}(A_{1ik\ell rs}^2, A_{1i'k\ell rs}^2),$$

with $A_{1ik\ell rs} = \mathbf{d}_{1kr}^\top \mathbf{d}_{i\ell s}$, $\mathbf{d}_{1kr} = \mathbf{x}_{1k} - \mathbf{x}_{1r}$, $\mathbf{d}_{i\ell s} = \mathbf{x}_{i\ell} - \mathbf{x}_{is}$, $i \in \{2, \dots, g\}$, as for two-sample case. The independence of samples, however, helps reduce this covariance to a simple form. First, by the trace properties, it follows that $E[(\mathbf{d}_{1kr}^\top \mathbf{d}_{i\ell s})^2 (\mathbf{d}_{1kr}^\top \mathbf{d}_{i'\ell s})^2] = 4E(\mathbf{d}_{1kr}^\top \Sigma_i \mathbf{d}_{1kr} \mathbf{d}_{i'\ell s}^\top \Sigma_{i'} \mathbf{d}_{i'\ell s})$, which can be computed using the results of Lemma 2. Ignoring, for convenience, the terms involving M 's which eventually vanish anyway under the assumptions (or exactly under normality), this expectation can be shown to be $32 \text{tr}(\Sigma_1 \Sigma_i \Sigma_{i'} \Sigma_{i'})$, so that

$$\text{Cov}(E_{1i}, E_{1i'}) = \frac{32}{n_1(n_1 - 1)} \text{tr}(\Sigma_1 \Sigma_i \Sigma_{i'} \Sigma_{i'}).$$

For $\text{Cov}(T_i, T_{i'})$, $i, i' = \{2, \dots, g\}$, $i \neq i'$, this leads to terms like

$$\frac{\text{tr}(\Sigma_1 \Sigma_i \Sigma_{i'} \Sigma_{i'})}{\sqrt{\|\Sigma_1\|^2 \|\Sigma_i\|^2} \sqrt{\|\Sigma_1\|^2 \|\Sigma_{i'}\|^2}} O\left(\frac{1}{n_1^2}\right),$$

which, by a simple application of Cauchy–Schwarz inequality, vanish under the assumptions, leaving the covariance matrix of $\mathbf{T}$, in the limit, a diagonal matrix. To express it precisely, along with the mean of $\mathbf{T}$, define $\mathbf{T}_0 = (T_{02}, \dots, T_{0g})^\top = (\sqrt{n_1 n_2} T_2, \dots, \sqrt{n_1 n_g} T_g)^\top$. Then $E(\mathbf{T}_0)$ and $\Omega = \text{Cov}(\mathbf{T}_0)$ can be approximated, for $n_i, p \rightarrow \infty$, along the lines of the two-sample case, as

$$E(\mathbf{T}_0) = \tau[1 + o(1)], \quad (20)$$

$$\text{Cov}(\mathbf{T}_0) = 4 \bigoplus_{i=2}^g (\tau_i^2 + 1)[1 + o(1)], \quad (21)$$

where $\tau = (\tau_2, \dots, \tau_g)^\top$ with $\tau_i = \|\Delta_1 \Delta_i\|^2 / \|\Sigma_1\| \|\Sigma_i\| - 1$, which is 0 under H_0 whence (20)–(21) reduce, respectively, to $\mathbf{0}$ and $4\mathbf{I}$, and $\bigoplus$ denotes Kronecker sum. For the joint distribution of $\mathbf{T}_0$ now, consider $T_g = \mathbf{1}^\top \mathbf{T}_0$ with $\mathbf{1}_{g-1}$ a vector of 1's, so that $E(T_g) = \mathbf{1}^\top E(\mathbf{T}_0) = \sum_{i=2}^g E(T_{0i})$ and $\text{Var}(T_g) = \mathbf{1}^\top \Omega \mathbf{1} = \sum_{i=2}^g \text{Var}(T_{0i})$. Following the proof of T in B, the joint distribution of $\mathbf{T}$ is, asymptotically a multivariate normal, by a simple application of the Cramér–Wold device. We state this in the following theorem.

Theorem 3. For $\mathbf{T}_0$ defined above, we have, under Model (7) and Assumptions 1–4,

$$\Omega^{-1}(\mathbf{T}_0 - E(\mathbf{T}_0)) \xrightarrow{\mathcal{D}} N_{g-1}(\mathbf{0}, 4\mathbf{I}), \quad (22)$$

as $n_i, p \rightarrow \infty$, $i \in \{2, \dots, g\}$, with $E(\mathbf{T}_0)$ and Ω given in (20)–(21), where $\mathbf{0}$ is a $(g-1) \times 1$ vector of zeros, and $\mathbf{I}$ is identity matrix of size $(g-1) \times (g-1)$. In particular, under H_0 , $\mathbf{T}_0 \xrightarrow{\mathcal{D}} N_{g-1}(\mathbf{0}, \mathbf{I})$.

Note that, as in two-sample case, the terms τ_i in Ω can be estimated using the estimators in Definition 1, so that the result in Theorem 3 remains valid by plugging in these consistent estimators.

4. Simulations

We evaluate T for its size control and power properties. We also compare it with [29]'s and [9]'s statistics, T_{TM} and T_{CLTZ} , as given in (4) and (6), respectively. For this, we take the same simulation parameters as in [29], including normality, except using one additional case for size and two additional cases for power. Apart from normal case, we also evaluate the three tests for t - and uniform distributions, to assess their robustness to normality and possible use under Model (7). We proceed as below.

Following [29], we define two parameter settings, each with several subcases. Define $d_1 \in \{3, 4, 5, 6, 7, 8, 9, 10\}$ and $d_2 \in \{2, 3, 4\}$. We generate n_i iid vectors $\mathbf{x}_{ik}$, $k \in \{1, \dots, n_i\}$, from certain p -dimensional non-degenerate distribution $\mathcal{F}_i$, where n_i and p differ in the two cases as:

- Case (I) $p = d_1^3$, $n_1 = n_2 = d_1^2$,
- Case (II) $p = d_2^5$, $n_1 = n_2 = d_2^3$.

Table 1

Size and power of T_{TM} , T_{CLTZ} and T in (4), (6) and (8), at $\alpha = 0.05$, for normal distribution under Cases I–II, where (A), ..., (N2) are defined in Section 4.

Case I	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
3	T	0.049	0.054	0.048	0.977	0.989	0.982	0.988	0.979	0.400	0.961
	T_{TM}	0.035	0.036	0.035	0.068	0.109	0.170	0.126	0.225	0.923	0.290
	T_{CLTZ}	0.051	0.050	0.032	0.084	0.111	0.155	0.112	0.203	0.819	0.270
4	T	0.053	0.046	0.051	0.993	1.000	1.000	1.000	0.993	0.783	1.000
	T_{TM}	0.040	0.038	0.039	0.117	0.264	0.460	0.325	0.615	1.000	0.767
	T_{CLTZ}	0.045	0.047	0.049	0.127	0.252	0.401	0.314	0.579	1.000	0.708
5	T	0.054	0.058	0.049	1.000	1.000	1.000	1.000	1.000	0.937	1.000
	T_{TM}	0.049	0.045	0.046	0.210	0.523	0.816	0.624	0.942	1.000	0.989
	T_{CLTZ}	0.056	0.046	0.029	0.184	0.513	0.792	0.628	0.927	1.000	0.986
6	T	0.050	0.052	0.047	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.051	0.048	0.054	0.339	0.812	0.986	0.900	0.999	1.000	1.000
	T_{CLTZ}	0.061	0.037	0.041	0.354	0.823	0.971	0.910	0.999	1.000	1.000
7	T	0.048	0.046	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.049	0.048	0.044	0.533	0.966	0.999	0.993	1.000	1.000	1.000
	T_{CLTZ}	0.052	0.048	0.052	0.556	0.965	1.000	0.993	1.000	1.000	1.000
8	T	0.047	0.055	0.056	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.046	0.048	0.040	0.720	0.999	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.050	0.051	0.046	0.733	1.000	1.000	1.000	1.000	1.000	1.000
9	T	0.054	0.051	0.054	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.047	0.046	0.045	0.887	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.054	0.048	0.044	0.893	1.000	1.000	1.000	1.000	1.000	1.000
10	T	0.051	0.053	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.049	0.049	0.044	0.969	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.051	0.047	0.053	0.967	1.000	1.000	1.000	1.000	1.000	1.000
Case II	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
2	T	0.057	0.049	0.053	0.974	0.971	0.959	0.961	0.938	0.356	0.925
	T_{TM}	0.037	0.033	0.030	0.057	0.092	0.137	0.102	0.184	0.903	0.249
	T_{CLTZ}	0.061	0.072	0.039	0.092	0.120	0.178	0.143	0.216	0.775	0.269
3	T	0.055	0.048	0.048	0.995	0.989	0.980	0.991	0.974	0.821	1.000
	T_{TM}	0.042	0.047	0.040	0.229	0.592	0.879	0.700	0.974	1.000	0.993
	T_{CLTZ}	0.041	0.040	0.056	0.248	0.628	0.865	0.722	0.974	1.000	0.996
4	T	0.053	0.048	0.051	1.000	1.000	0.999	1.000	0.989	1.000	1.000
	T_{TM}	0.051	0.049	0.047	0.729	0.998	1.000	0.999	1.000	1.000	1.000
	T_{CLTZ}	0.048	0.055	0.051	0.733	0.998	1.000	1.000	1.000	1.000	1.000

The triplets (n_1, n_2, p) in each case are combined with the true covariance structure, where Σ_1 is always assumed an identity matrix, $\mathbf{I}$ of order p , except case (N) for size and (N2) for power (see below), and Σ_2 is assumed to follow three structures for size and seven structures for power. These structures are listed below as (A)–(N) for size and (C)–(N2) for power, where $\mathbf{1}$ denotes a $p \times 1$ vector of 1s:

- (A) $\Sigma_2 = \mathbf{I}_p$;
- (B) $\Sigma_2 = 5\mathbf{I}_p$;
- (N) $(\Sigma_1)_{jj'} = 0.5^{|i-j|}$, $\Sigma_2 = 5\Sigma_1$;
- (C) $(\Sigma_2)_{jj'} = \{1 + 4(i-1)/(p-1)\} \times \mathbf{1}(i=j)$;
- (D) $(\Sigma_2)_{jj'} = \mathbf{1}(i=j) + 0.4 \times \mathbf{1}(|i-j|=1)$;
- (E) $(\Sigma_2)_{jj'} = \mathbf{1}(i=j) + 0.5 \times \mathbf{1}(|i-j|=1)$;
- (F) $(\Sigma_2)_{jj'} = 0.4^{|i-j|}$;
- (G) $(\Sigma_2)_{jj'} = 0.5^{|i-j|}$;
- (N1) $(\Sigma_2)_{jj'} = (1-0.5)\mathbf{I}_p + 0.5\mathbf{1}\mathbf{1}^T$;
- (N2) Σ_1 as in (C), Σ_2 as in (G).

Here, $(\Sigma)_{jj'}$ denotes (j, j') th entry in the matrix, $j, j' \in \{1, \dots, p\}$ and $\mathbf{1}$ denotes a vector of 1's. Note that, (F) and (G) represent autoregressive structures of order 1 with $\rho = 0.4$ and 0.5 , respectively, where (D) and (E) are their extended versions. Further, (N1) is a compound symmetric structure, $\kappa(1-\rho)\mathbf{I}_p + \rho\mathbf{J}_p$ with $\kappa = 1$, $\rho = 0.5$, where $\mathbf{J} = \mathbf{1}\mathbf{1}^T$ is a matrix of 1's. The subcases (A)–(B) for size and (C)–(G) for power are the same as used in [29], where subcase (N) for size and subcases (N1)–(N2) for power are new, to assess the tests under a wider variety of parameter settings. We use nominal $\alpha = 0.05$ for both size and power.

We evaluate the tests for normal, t_{10} , t_5 and uniform distributions, the results of which are reported in Tables 1–4, respectively. The upper panel of each table is reserved for Case I and lower panel for Case II, where the columns represent

Table 2Size and power of T_{TM} , T_{CLTZ} and T in (4), (6) and (8), at $\alpha = 0.05$, for t_{10} distribution under Cases I–II, where (A), ..., (N2) are defined in Section 4.

Case I	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
3	T	0.058	0.051	0.057	0.885	0.969	0.994	0.996	0.994	0.745	0.991
	T_{TM}	0.032	0.030	0.038	0.724	0.871	0.874	0.885	0.891	0.873	0.905
	T_{CLTZ}	0.050	0.044	0.035	0.127	0.109	0.153	0.125	0.211	0.816	0.250
4	T	0.047	0.050	0.048	0.998	1.000	1.000	1.000	1.000	0.994	1.000
	T_{TM}	0.034	0.048	0.044	0.801	0.997	0.999	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.053	0.048	0.036	0.205	0.252	0.405	0.318	0.556	0.999	0.740
5	T	0.053	0.047	0.050	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.048	0.052	0.051	0.994	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.045	0.050	0.038	0.387	0.521	0.802	0.615	0.930	1.000	0.983
6	T	0.049	0.050	0.050	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.045	0.043	0.041	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.048	0.050	0.039	0.513	0.798	0.978	0.907	0.999	1.000	1.000
7	T	0.047	0.051	0.053	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.047	0.054	0.052	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.060	0.053	0.044	0.701	0.951	1.000	0.991	1.000	1.000	1.000
8	T	0.053	0.051	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.045	0.040	0.042	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.049	0.058	0.038	0.896	0.999	1.000	1.000	1.000	1.000	1.000
9	T	0.051	0.048	0.052	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.054	0.041	0.044	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.054	0.051	0.045	0.974	1.000	1.000	1.000	1.000	1.000	1.000
10	T	0.049	0.052	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.055	0.047	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.048	0.052	0.047	0.998	1.000	1.000	1.000	1.000	1.000	1.000
Case II	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
2	T	0.059	0.057	0.057	0.991	0.987	0.983	0.988	0.985	0.674	0.978
	T_{TM}	0.041	0.037	0.039	0.815	0.881	0.872	0.869	0.887	0.801	0.894
	T_{CLTZ}	0.079	0.070	0.046	0.079	0.103	0.158	0.142	0.203	0.766	0.240
3	T	0.048	0.051	0.046	1.000	1.000	1.000	1.000	1.000	0.953	1.000
	T_{TM}	0.036	0.042	0.040	1.000	1.000	1.000	1.000	1.000	0.988	1.000
	T_{CLTZ}	0.063	0.055	0.043	0.246	0.600	0.888	0.730	0.972	1.000	1.000
4	T	0.052	0.045	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.053	0.044	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.057	0.051	0.046	0.745	0.998	1.000	1.000	1.000	1.000	1.000

the subcases, with first three columns for size and next seven for power. All results are averages over 3000 simulation runs, except those for T_{TM} in Table 1 which are directly taken from Tables 1–2 in [29].

To begin with, we observe an accurate performance of T , for both size and power, and in both cases with all their subcases. In particular, even for the smallest sample sizes, the size is highly accurately estimated and the test attains a very high power. Further, this trend remains intact for increasing dimension, and also for all distributions. The accuracy of T for t - and uniform distributions shows robustness of T to normality and applicability under general model like (7).

The performance of T_{TM} under normality is very close to that of T , except for small sample sizes whence it is conservative in size estimation and has low power. However, with increasing sample size, it improves reasonably well, and this holds for both cases. A similar trend can also be witnessed for t_{10} and t_5 distributions. A particular feature, for new parameter sets, is worth mentioning. Whereas size estimation under (N) mimics the same trend as under other subcases, the power of the two tests show opposite trends, favoring T_{TM} and T , respectively, under (N1) and (N2). However, both tests again approach the maximum power quickly, for a reasonable sample size. Moreover, this differential feature of the two tests remains same for all distributions.

For uniform distribution, however, the performance of T_{TM} seems sub-optimal. The size is consistently highly conservatively estimated, mostly around 0.01–0.02 for nominal size of 0.05. Further, the power is low for small sample size, for all subcases, and grows relatively slowly as compared to that under normal and t distributions, so that the test needs reasonably large sample sizes to gain good power. The size performance, however, does not improve, even for increasing sample size. Finally, this trend is likewise mimicked in Case II as well. Combined with the result for t -distribution, it can be concluded that the robustness of T_{TM} is limited to certain close cases, e.g., for elliptically contoured distributions to which both normal and t distributions belong.

The statistic T_{CLTZ} appears to perform more accurately, and more closely to T , although there are also exceptions in this case. First, it also starts conservative with small sample sizes but improves very quickly, for both normal and non-normal cases. For power, it generally outperforms T_{TM} and gains maximum power rather quickly. Its highly accurate size performance, also for t -distribution, is particularly remarkable, although it needs relatively larger sample sizes to gain high power for t -distribution. For uniform distribution, however, it depicts opposite exceptions compared to T_{TM} , i.e., it

Table 3Size and power of T_{TM} , T_{CLTZ} and T in (4), (6) and (8), at $\alpha = 0.05$, for t_5 distribution under Cases I–II, where (A), ..., (N2) are defined in Section 4.

Case I	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
3	T	0.051	0.040	0.047	0.935	0.997	0.984	0.995	0.996	0.908	0.998
	T_{TM}	0.033	0.036	0.042	0.842	0.860	0.874	0.855	0.889	0.989	0.894
	T_{CLTZ}	0.038	0.052	0.027	0.085	0.110	0.170	0.130	0.217	0.818	0.711
4	T	0.046	0.055	0.048	1.000	1.000	1.000	1.000	1.000	0.999	1.000
	T_{TM}	0.036	0.041	0.040	1.000	1.000	1.000	1.000	1.000	1.000	0.998
	T_{CLTZ}	0.066	0.059	0.041	0.125	0.263	0.444	0.296	0.583	1.000	0.984
5	T	0.048	0.046	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.040	0.043	0.044	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.041	0.061	0.048	0.212	0.525	0.809	0.617	0.919	1.000	1.000
6	T	0.054	0.046	0.047	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.038	0.045	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.045	0.052	0.042	0.334	0.775	0.982	0.906	0.999	1.000	1.000
7	T	0.052	0.050	0.045	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.049	0.046	0.050	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.048	0.053	0.043	0.526	0.968	0.999	0.995	1.000	1.000	1.000
8	T	0.056	0.047	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.051	0.045	0.047	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.044	0.051	0.047	0.728	0.996	1.000	1.000	1.000	1.000	1.000
9	T	0.048	0.047	0.055	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.050	0.053	0.046	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.046	0.052	0.048	0.885	1.000	1.000	1.000	1.000	1.000	1.000
10	T	0.051	0.048	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.053	0.054	0.056	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.044	0.050	0.046	0.979	1.000	1.000	1.000	1.000	1.000	1.000
Case II	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
2	T	0.047	0.054	0.049	0.981	0.990	0.989	0.990	0.993	0.869	0.990
	T_{TM}	0.034	0.039	0.041	0.854	0.861	0.858	0.877	0.890	0.983	0.906
	T_{CLTZ}	0.057	0.063	0.041	0.09	0.127	0.156	0.150	0.183	0.757	0.244
3	T	0.049	0.047	0.052	1.000	1.000	1.000	1.000	1.000	0.947	1.000
	T_{TM}	0.038	0.044	0.047	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.045	0.041	0.044	0.249	0.566	0.869	0.732	0.975	1.000	0.994
4	T	0.055	0.051	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.047	0.046	0.044	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.053	0.045	0.048	0.750	0.999	1.000	1.000	1.000	1.000	1.000

is liberal for small sample sizes. But it improves quickly and discernibly. Finally, these properties of T_{CLTZ} remain mostly intact for both cases and their subcases.

In short, it can be concluded that the three tests perform well under normality and under high-dimensional set up, where sample sizes can be moderate. Further, the tests also show their robustness, with T being generally robust, followed closely by T_{CLTZ} , whereas T_{TM} has its limitations for distributions which are relatively far from normal such as uniform.

5. Discussion and conclusions

A test statistic is proposed for testing proportionality of two large covariance matrices. Using an equivalence formulation of the null hypothesis, which helps dispense with the proportionality constant, the test is composed of simple, computationally efficient, and consistent estimators. Simulations show the accuracy of the test for a variety of parameter settings, particularly emphasizing high-dimensionality and robustness to normality. A multi-sample extension is also given.

The test statistic is also compared with its two most recent competitors, constructed along similar lines. The proposed test generally outperforms both in most cases, with T_{CLTZ} a closer competitor, particularly in size accuracy. It may, however, be mentioned that computationally, T_{CLTZ} is more expensive than T and T_{TM} , in the sense of its moments, even under the null. In this context, T is the simplest in its construction and moments and has highly accurate performance.

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Table 4

Size and power of T_{TM} , T_{CLTZ} and T in (4), (6) and (8), at $\alpha = 0.05$, for uniform distribution under Cases I–II, where (A), ..., (N2) are defined in Section 4.

Case I	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
3	T	0.045	0.042	0.045	0.728	0.889	0.996	0.992	0.995	0.510	0.441
	T_{TM}	0.011	0.013	0.014	0.014	0.041	0.070	0.049	0.089	0.600	0.111
	T_{CLTZ}	0.080	0.072	0.074	0.071	0.163	0.216	0.152	0.252	0.700	0.264
4	T	0.044	0.049	0.047	0.969	1.000	1.000	1.000	1.000	0.854	0.803
	T_{TM}	0.017	0.014	0.015	0.012	0.127	0.257	0.166	0.396	0.987	0.389
	T_{CLTZ}	0.067	0.050	0.059	0.061	0.273	0.456	0.314	0.609	0.991	0.605
5	T	0.049	0.046	0.044	1.000	1.000	1.000	1.000	1.000	1.000	0.996
	T_{TM}	0.019	0.011	0.012	0.013	0.279	0.579	0.349	0.802	1.000	0.798
	T_{CLTZ}	0.062	0.060	0.065	0.060	0.498	0.781	0.592	0.921	1.000	0.920
6	T	0.050	0.048	0.052	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.017	0.010	0.014	0.015	0.585	0.917	0.687	0.990	1.000	0.995
	T_{CLTZ}	0.070	0.059	0.054	0.051	0.789	0.974	0.874	0.995	1.000	0.996
7	T	0.049	0.048	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.019	0.014	0.0110	0.016	0.861	0.989	0.806	1.000	1.000	1.000
	T_{CLTZ}	0.042	0.043	0.054	0.063	0.951	0.999	0.987	1.000	1.000	1.000
8	T	0.048	0.050	0.047	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.015	0.011	0.012	0.011	0.951	1.000	0.910	1.000	1.000	1.000
	T_{CLTZ}	0.044	0.050	0.063	0.060	0.999	1.000	1.000	1.000	1.000	1.000
9	T	0.051	0.050	0.049	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.017	0.008	0.011	0.012	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.046	0.053	0.060	0.055	1.000	1.000	1.000	1.000	1.000	1.000
10	T	0.050	0.049	0.051	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	T_{TM}	0.013	0.008	0.012	0.013	1.000	1.000	1.000	1.000	1.000	1.000
	T_{CLTZ}	0.047	0.054	0.058	0.052	1.000	1.000	1.000	1.000	1.000	1.000
Case II	Statistic	Size			Power						
		(A)	(B)	(N)	(C)	(D)	(E)	(F)	(G)	(N1)	(N2)
2	T	0.049	0.056	0.045	0.836	0.984	0.981	0.986	0.972	0.364	0.753
	T_{TM}	0.011	0.012	0.015	0.009	0.037	0.053	0.049	0.099	0.536	0.092
	T_{CLTZ}	0.113	0.11	0.097	0.099	0.186	0.226	0.188	0.255	0.664	0.277
3	T	0.053	0.052	0.046	1.000	1.000	1.000	1.000	1.000	0.741	0.980
	T_{TM}	0.006	0.009	0.011	0.009	0.318	0.631	0.409	0.887	1.000	0.892
	T_{CLTZ}	0.064	0.066	0.053	0.059	0.580	0.878	0.713	0.973	1.000	0.956
4	T	0.046	0.046	0.0551	1.000	1.000	1.000	1.000	1.000	0.996	1.000
	T_{TM}	0.004	0.012	0.011	0.006	0.502	0.799	0.571	0.999	1.000	1.000
	T_{CLTZ}	0.059	0.056	0.049	0.051	0.742	0.904	0.886	1.000	1.000	1.000

Appendix A. Some basic results

Given $\mathbf{y}_{ik}$ and $\mathbf{z}_{ik}$ as in (7), define

$$A_{ik} = \mathbf{y}_{ik}^\top \mathbf{y}_{ik} = \mathbf{z}_{ik}^\top \Sigma_i \mathbf{z}_{ik}, \quad A_{ikr} = \mathbf{y}_{ik}^\top \mathbf{y}_{ir} = \mathbf{z}_{ik}^\top \Sigma_i \mathbf{z}_{ir}, \quad A_{ijk\ell} = \mathbf{y}_{ik}^\top \mathbf{y}_{j\ell} = \mathbf{z}_{ik}^\top \Delta_i \Delta_j \mathbf{z}_{j\ell},$$

as, respectively, a quadratic form of vectors from i th sample, a symmetric bilinear form of independent vectors from i th sample, $k \neq r$, $k, r \in \{1, \dots, n_i\}$, and a symmetric bilinear form of independent vectors from i th and j th samples, $i \neq j$, $i, j \in \{1, \dots, g\}$, $g \geq 2$. To study the properties of T , we need moments of these quadratic and bilinear forms. These moments involve, under Model (7), certain terms. As all these terms eventually vanish, exactly under normality and asymptotically under the assumptions, we denote them, for simplicity, using a general matrix $\mathbf{A} = (a_{st})_{s,t=1}^p$, in the following, where $\odot$ denotes the Hadamard product.

$$M_1 = \gamma \operatorname{tr}(\mathbf{A} \odot \mathbf{A}), \quad M_2 = \gamma \operatorname{tr}(\mathbf{A}^2 \odot \mathbf{A}^2), \quad M_3 = 6\gamma \operatorname{tr}(\mathbf{A}^2 \odot \mathbf{A}^2) + \gamma^2 \sum_{s=1}^p \sum_{t=1}^p a_{st}^4,$$

$$M_4 = 2\gamma \operatorname{tr}(\mathbf{A}) \operatorname{tr}(\mathbf{A}^2 \odot \mathbf{A}) + 4\gamma \operatorname{tr}(\mathbf{A}^3 \odot \mathbf{A}) + \gamma \operatorname{tr}(\mathbf{A} \odot \mathbf{A} \mathbf{d} \mathbf{A}), \quad \mathbf{d} = \operatorname{diag}(\mathbf{A}).$$

Theorem 4. For A_{ik} , A_{ikr} , $A_{ijk\ell}$ given above, we have $E(A_{ik}^2) = 2 \operatorname{tr}(\Sigma_i^2) + \{\operatorname{tr}(\Sigma_i)\}^2 + M_2$, $E(A_{ijk\ell}^4) = 6 \operatorname{tr}(\Sigma_i \Sigma_j)^2 + 3\{\operatorname{tr}(\Sigma_i \Sigma_j)\}^2 + M_3$, $E(A_{ik} A_{jk}) = 2 \operatorname{tr}(\Sigma_i^3 \Sigma_j) + \operatorname{tr}(\Sigma_i^2) \operatorname{tr}(\Sigma_j \Sigma_j) + M_2$, $E(Q_{ik} Q_{jk} A_{ijk\ell}^2) = 4 \operatorname{tr}(\Sigma_i \Sigma_j)^2 + 4 \operatorname{tr}(\Sigma_i^3) \operatorname{tr}(\Sigma_j) + \{\operatorname{tr}(\Sigma_i)\}^2 \operatorname{tr}(\Sigma_j^2) + M_4$, with $M_1, \dots, M_4$ given above. Further, $E(A_{ik}) = \operatorname{tr}(\Sigma_i)$, $E(A_{ikr}^2) = \operatorname{tr}(\Sigma_i^2)$, $\operatorname{Cov}(A_{ik}, A_{ikr}) = 0$.

Following extended moments can be computed using Theorem 4; see e.g., [1].

Lemma 2. From Theorem 4, we have the following for $\mathbf{y}_{ik}$ as in Model (7):

$$\begin{aligned} E(\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{i\ell}^\top \Sigma_i \mathbf{y}_{ir}) &= \text{tr}(\Sigma_i^4) = E(\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{i\ell}^\top \mathbf{y}_{ir} \mathbf{y}_{i\ell}^\top \Sigma_i \mathbf{y}_{ir}); \\ \text{Cov}((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2, \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{ik}) &= 2 \text{tr}(\Sigma_i^3 \Sigma_j) + M_2; \\ E((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2 \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{ik}) &= 2 \text{tr}(\Sigma_i^4) + (\text{tr}(\Sigma_i^2))^2 + M_2 = \text{Var}(\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{i\ell}); \\ E((\mathbf{y}_{i\ell}^\top \mathbf{y}_{ir})^2 \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{ik}) &= \text{tr}(\Sigma_i \Sigma_j) \text{tr}(\Sigma_i)^2; \\ E((\mathbf{y}_{ik}^\top \mathbf{y}_{ju})^2 \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{ik}) &= 2 \text{tr}((\Sigma_i \Sigma_j)^2) + (\text{tr}(\Sigma_i \Sigma_j))^2 + M_2 = \text{Var}(\mathbf{y}_{ik}^\top \mathbf{y}_{ju} \mathbf{y}_{ir}^\top \mathbf{y}_{ju}); \\ \text{Cov}((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2, (\mathbf{y}_{ik}^\top \mathbf{y}_{ir})^2) &= 2 \text{tr}(\Sigma_i^4) + M_2; \\ E(\mathbf{y}_{jk}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{ir}) &= \text{tr}((\Sigma_i \Sigma_j)^2) = \text{Cov}(\mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{i\ell}, \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{i\ell}); \\ \text{Cov}((\mathbf{y}_{jk}^\top \mathbf{y}_{i\ell})^2, (\mathbf{y}_{jk}^\top \mathbf{y}_{ir})^2) &= 2 \text{tr}((\Sigma_i \Sigma_j)^2) + M_2. \end{aligned}$$

Furthermore, $E((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2 \mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{i\ell}^\top \Sigma_i \mathbf{y}_{ir})$, $E(\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{i\ell}^\top \Sigma_i \mathbf{y}_{ik})$, $E(\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell} \mathbf{y}_{ir}^\top \mathbf{y}_{i\ell}^\top \Sigma_i \mathbf{y}_{i\ell})$, $E((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2 \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{i\ell})$, $E((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2 \mathbf{y}_{ik}^\top \Sigma_j \mathbf{y}_{i\ell})$, and $E((\mathbf{y}_{ik}^\top \mathbf{y}_{i\ell})^2 \mathbf{y}_{ik}^\top \mathbf{y}_{ir} \mathbf{y}_{i\ell}^\top \mathbf{y}_{ir})$ all vanish.

Following special version of Lindeberg–Feller central limit theorem, also known as Hájek–Šidák Lemma, will be used to fix the limit of the test statistic, see B.2. below. For details, see e.g., [22] or [18, p 184]; see also [4] for a similar application of this lemma.

Lemma 3. Let $X_1, X_2, \dots$ be iid random variables with mean 0 and variance 1. Let b_{ni} , $1 \leq i \leq n$, be constants such that $\max_i b_{ni}^2 \rightarrow 0$ as $n \rightarrow \infty$. Then $\sum_{i=1}^n b_{ni} X_i \xrightarrow{D} N(0, 1)$, as $n \rightarrow \infty$.

Appendix B. Main proofs

Proof of Theorem 1. The estimator $\hat{\Sigma}_i$ in (2) can be equivalently written as $\hat{\Sigma}_i = [1/n_i(n_i - 1)] \sum_{k \neq r}^{n_i} \mathbf{d}_{ikr} \mathbf{d}_{ikr}^\top / 2$ with $\mathbf{d}_{ikr} = \mathbf{y}_{ik} - \mathbf{y}_{ir}$ (see (7)), where $E(\mathbf{d}_{ikr}) = 0$, $\text{Cov}(\mathbf{d}_{ikr}) = E(\mathbf{d}_{ikr} \mathbf{d}_{ikr}^\top) = 2 \Sigma_i$, $i \in \{1, 2\}$. This leads to $E_{12} = \text{tr}(\hat{\Sigma}_1 \hat{\Sigma}_2)$ as a two-sample U -statistic

$$E_{12} = \frac{1}{n_1 n_2 (n_1 - 1)(n_2 - 1)} \sum_{\substack{k=1 \\ k \neq r}}^{n_1} \sum_{\substack{r=1 \\ r \neq k}}^{n_1} \sum_{\substack{\ell=1 \\ \ell \neq s}}^{n_2} \sum_{\substack{s=1 \\ s \neq \ell}}^{n_2} \frac{1}{4} A_{12k\ell rs}^2,$$

with symmetric kernel $A_{12k\ell rs}^2/4$, where $A_{12k\ell rs} = \mathbf{d}_{1kr}^\top \mathbf{d}_{2\ell s}$. By the same argument, we also have

$$E_i = \frac{1}{P(n_i)} \sum_{k=1}^n \sum_{\ell=1}^n \sum_{r=1}^n \sum_{s=1}^n B_{ikr\ell s},$$

$\pi(k, \ell, r, s)$

as one-sample U -statistic of symmetric kernel, $B_{ikr\ell s}/12$, $B_{ikr\ell s} = A_{i\ell s k r}^2 + A_{i\ell k s r}^2 + A_{i\ell r s k}^2$, $A_{i\ell s k r} = \mathbf{d}_{i\ell s}^\top \mathbf{d}_{ikr}$, $P(n_i) = n_i(n_i - 1)(n_i - 2)(n_i - 3)$ and $\pi(k, \ell, r, s)$ implies all indices pairwise unequal. With $E(A_{i\ell s k r}^2) = 4 \|\Sigma_i\|^2$, it also gives the unbiasedness part of Theorem 1.

The variances follow from the conditional expectations (projections) of the respective kernels; see e.g., [28]. For $U_{n_1 n_2}$, with kernel $h(k, \ell, r, s) = A_{12k\ell rs}^2/4$, as defined above, we obtain

$$\begin{aligned} h_{10} &= [\mathbf{y}_{1k}^\top \Sigma_2 \mathbf{y}_{1k} + \text{tr}(\Sigma_1 \Sigma_2)]/2, \quad h_{01} = [\mathbf{y}_{2\ell}^\top \Sigma_1 \mathbf{y}_{2\ell} + \text{tr}(\Sigma_1 \Sigma_2)]/2, \quad h_{20} = (\mathbf{d}_{1kr}^\top \Sigma_2 \mathbf{d}_{1kr})/4, \\ h_{02} &= (\mathbf{d}_{2\ell s}^\top \Sigma_1 \mathbf{d}_{2\ell s})/4, \quad h_{11} = [(\mathbf{y}_{1k}^\top \mathbf{y}_{2\ell})^2 + \mathbf{y}_{1k}^\top \Sigma_2 \mathbf{y}_{1k} + \mathbf{y}_{2\ell}^\top \Sigma_1 \mathbf{y}_{2\ell} + \text{tr}(\Sigma_1 \Sigma_2)]/4, \\ h_{21} &= [(\mathbf{d}_{1kr}^\top (\mathbf{y}_{2\ell} \mathbf{y}_{2\ell}^\top + \Sigma_2) \mathbf{d}_{1kr})]/4, \quad h_{12} = [(\mathbf{d}_{2\ell s}^\top (\mathbf{y}_{1k} \mathbf{y}_{1k}^\top + \Sigma_1) \mathbf{d}_{2\ell s})]/4, \end{aligned}$$

with $h_{22}(\cdot) = h(\cdot)$, where $h_{c_1 c_2} = E[h(k, \ell, r, s) | \mathbf{y}_{11}, \dots, \mathbf{y}_{ij}, j \in \{1, \dots, c_i\}, c_i \in \{0, 1, 2\}, i \in \{1, 2\}, h_{00} = 0$. The variances of these projections follow from Theorem 4 and Lemma 2 which, on substitution, give $\text{Var}(E_{12})$. We avoid detailed arithmetics which can be found in [1,2]. For E_i with $h_i(k, \ell, r, s) = B_{ikr\ell s}/12$, the projections, $E[h(\cdot) | \mathbf{y}_{11}, \dots, \mathbf{y}_{ic_i}]$, $c_i \in \{1, \dots, 4\}$, $i \in \{1, 2\}$, follow as

$$\begin{aligned} h_{i1} &= [6 \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{ik} + 6 \text{tr}(\Sigma_i^2)]/12, \\ h_{i2} &= [4 \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{ik} + 4 \mathbf{y}_{ir}^\top \Sigma_i \mathbf{y}_{ir} + 2(\mathbf{y}_{ik}^\top \mathbf{y}_{ir})^2 - 4 \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{ir} + 2 \text{tr}(\Sigma_i^2)]/12, \\ h_{i3} &= 2[(\mathbf{y}_{ik}^\top \mathbf{y}_{ik'})^2 + \mathbf{y}_{ik}^\top \Sigma_i \mathbf{y}_{ik} + (\mathbf{y}_{ir}^\top \mathbf{y}_{ik'})^2 + \mathbf{y}_{ir}^\top \Sigma_i \mathbf{y}_{ir} - \mathbf{y}_{ik}^\top \mathbf{y}_{ik'} \mathbf{y}_{ir}^\top \mathbf{y}_{ik'} - \mathbf{y}_{ik}^\top \mathbf{y}_{ir} \mathbf{y}_{ik'}^\top \mathbf{y}_{ir}]/12, \end{aligned}$$

and $h_{i4} = h_i(\cdot)$, with their variances obtained again, after tedious computations, using the moments in Theorem 4 and Lemma 2. The covariance term is relatively easy, and follows similarly from the covariance formula between two U -statistics; see [2].

Proof of Theorem 2. The proof follows by a sequence of steps. First, by the consistency of E_i/p^2 , discussed after Theorem 1, it suffices to focus on T as in (14), which in turn implies that we essentially need the limit of E_{12}/p^2 . From Theorem 1, it appears that $\sqrt{n_1 n_2} E_{12}$ might have a non-degenerate limit, under the assumptions, as $n_i \rightarrow \infty$. However, with $p \rightarrow \infty$, E_{12} has a varying kernel.

Recently, the theory of U -statistics has got special attraction for high-dimensional inference. The key point to be taken care of is the behavior of the variances of projection kernels (see the proof above). For details in other similar contexts, see [1,3,4] and the references therein.

By the general theory of U -statistics, a normal limit usually follows from first-order (Hájék) projections given that the corresponding variances are positive. As we see in the sequel, this is not so in the present case. Rather, an extended limit, using second-order projections, is needed, which coincides with the intuitive idea of $\sqrt{n_1 n_2} E_{12}$ having a non-degenerate limit.

To see this, let the two independent samples in E_{12} , $\mathbf{x}_{ik} \in \mathbb{R}^p$, $k \in \{1, \dots, n_i\}$, each composed of iid vectors following a distribution $\mathcal{F}_i$, $i \in \{1, 2\}$, be generated by a Borel space $(\mathcal{X}, \mathcal{B})$. The kernel, $h(k, r, \ell, s) = A_{12k\ell rs}^2/4$, $A_{12k\ell rs} = \mathbf{d}_{1kr}^\top \mathbf{d}_{2\ell s}$, $\mathbf{d}_{1kr} = \mathbf{y}_{1k} - \mathbf{y}_{1r}$, $\mathbf{y}_{1k} = \mathbf{x}_{1k} - \boldsymbol{\mu}_1$ etc., is a symmetric map, $h : (\mathcal{X}^{m_1+m_2}, \mathcal{B}^{m_1+m_2}) \rightarrow (\mathbb{R}, \mathbb{B})$, where m_i is the order of the kernel for i th sample. In our case, $m_i = 2$, $i \in \{1, 2\}$. Assuming $h(\cdot) \in \mathbb{L}_1(\mathcal{F}_2^2 \otimes \mathcal{F}_2^2)$, the unbiasedness of $h(\cdot)$ exists for the symmetric functional $\tau = \tau(\mathcal{F}_1, \mathcal{F}_2) = \|\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2\|^2$, i.e.,

$$\tau(\mathcal{F}_1, \mathcal{F}_2) = \int \dots \int h(k, r, \ell, s) \prod_{i=1}^2 \prod_{j=1}^{m_i} d\mathcal{F}_i(\mathbf{x}_{ij}),$$

where $h(k, r, \ell, s)$ denotes, for notational convenience, the kernel $h(\mathbf{x}_{11}, \dots, \mathbf{x}_{1n_1}; \mathbf{x}_{21}, \dots, \mathbf{x}_{2n_2})$.

Now, denote $\tilde{U}_{n_1 n_2} = \tilde{E}_{12} = E_{12} - E(E_{12})$ with kernel $\tilde{h}(\cdot) = h(\cdot) - E[h(\cdot)]$. From B.1., the first order projection of $\tilde{U}_{n_1 n_2}$ [see e.g., 30, p.165] follows as

$$\hat{U}_{n_1 n_2} = \frac{2}{n_1} \sum_{k=1}^{n_1} \tilde{h}_{10} + \frac{2}{n_2} \sum_{\ell=1}^{n_2} \tilde{h}_{01} = \frac{2}{n_1} \sum_{k=1}^{n_1} [\mathbf{y}_{1k}^\top \boldsymbol{\Sigma}_2 \mathbf{y}_{1k} - \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)] + \frac{2}{n_2} \sum_{\ell=1}^{n_2} [\mathbf{y}_{2\ell}^\top \boldsymbol{\Sigma}_1 \mathbf{y}_{2\ell} - \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)],$$

with $E(\hat{U}_{n_1 n_2}) = 0$. In fact, it can further be shown that $\text{Cov}(\tilde{U}_{n_1 n_2}, \hat{U}_{n_1 n_2}) = \text{Var}(\hat{U}_{n_1 n_2})$, so that $\text{Var}(\hat{U}_{n_1 n_2})/\text{Var}(\tilde{U}_{n_1 n_2}) \rightarrow 1$, implying that $\text{Cov}(\tilde{U}_{n_1 n_2}, \hat{U}_{n_1 n_2})/\text{Var}(\hat{U}_{n_1 n_2}) \rightarrow 1$, as essential conditions for normal limit; see e.g., [24, p.144]. We, however, note that $\text{Var}(\hat{U}_{n_1 n_2})$, composed of $\xi_{10} = \text{Var}(\tilde{h}_{10})$, $\xi_{01} = \text{Var}(\tilde{h}_{01})$, simplifies to $[2n/n_1 n_2] \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)^2$, $n = n_1 + n_2$, so that for E_{12}/p^2 , this variance vanishes as $p \rightarrow \infty$, leaving the kernel first-order degenerate in the limit. We thus need to consider the second-order projections.

Denote $\tilde{h}_{02} = h_{02} - E(h_{02})$, similarly $\tilde{h}_{20}, \tilde{h}_{11}$. From the proof above, $\xi_{02} = \xi_{20} = 2 \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)^2$ and $\xi_{11} = [9 \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)^2 + [\text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)^2]^2]/8$, with $\xi_{02} = \text{Var}(\tilde{h}_{02})$, $\xi_{11} = \text{Var}(\tilde{h}_{11})$. Thus, ξ_{02}, ξ_{20} , and the first term in ξ_{11} vanish, as before, and a non-degenerate limit for $p \rightarrow \infty$ is ensured only by the second term in ξ_{11} . To see how this limit follows precisely, let ν_{12s} , $s \in \{1, \dots, p\}$, be the eigenvalues of $\boldsymbol{\Sigma}_1^{1/2} \boldsymbol{\Sigma}_2^{1/2}/p = \boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2/p$, or of $\boldsymbol{\Delta}_2 \boldsymbol{\Delta}_1/p$, and write the kernel of E_{12} , ignoring 4 for simplicity, as

$$A_{12k\ell rs}^2/p^2 = (\mathbf{d}_{1kr}^\top \mathbf{d}_{2\ell s})(\mathbf{d}_{2\ell s}^\top \mathbf{d}_{1kr})/p^2 = (\mathbf{d}_{1kr}^\top \otimes \mathbf{d}_{2\ell s}^\top)(\mathbf{d}_{2\ell s} \otimes \mathbf{d}_{1kr})/p^2 = \text{tr}[\mathbf{d}_{2\ell s} \mathbf{d}_{1kr}^\top \otimes \mathbf{d}_{1kr} \mathbf{d}_{2\ell s}^\top]/p^2.$$

Since $E(\mathbf{d}_{1kr}^\top \mathbf{d}_{1kr}) = 0$ and $\text{Cov}(\mathbf{d}_{2\ell s} \mathbf{d}_{1kr}) = 4 \text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)$, it follows that $\theta_{12s} = \nu_{12s} \nu_{21s}$ denote the eigenvalues of $[(\boldsymbol{\Delta}_1 \boldsymbol{\Delta}_2) \otimes (\boldsymbol{\Delta}_2 \boldsymbol{\Delta}_1)]/p^2 = \boldsymbol{\Lambda}_{12}$ so that, $\sum_{s=1}^p \theta_{12s}^2 = \text{tr}(\boldsymbol{\Lambda}_{12}^2) = [\text{tr}(\boldsymbol{\Sigma}_1 \boldsymbol{\Sigma}_2)]^2/p^2$. Then, the weak convergence of $\sqrt{n_1 n_2} \tilde{E}_{12}$, $E_{12} = (E_{12} - E(E_{12}))/p^2$, for $n_i \rightarrow \infty$ is given as [see 31, Ch.4]

$$\tilde{E}_{12} \xrightarrow{\mathcal{D}} \sum_s \theta_{12s} Z_{1s} Z_{2s},$$

where Z_{is} , $i \in \{1, 2\}$, are two independent sequences of iid $N(0, 1)$ variables. To extend the limit to (14) and $p \rightarrow \infty$, let ϑ_{is} , $s \in \{1, \dots, p\}$, be the eigenvalues of $\boldsymbol{\Sigma}_i/p$, $i \in \{1, 2\}$, so that $\text{tr}(\boldsymbol{\Sigma}_i^2)/p^2 = \sum_{s=1}^p \vartheta_{is}^2$. Then, the limit above can be extended for (14) as

$$\sum_{s=1}^p c_s W_s,$$

with $c_s^2 = \theta_{12s}^2/(\sum_{s=1}^p \vartheta_{1s}^2 \sum_{s=1}^p \vartheta_{2s}^2)$ and $W_s = Z_{1s} Z_{2s}$. The normal limit of $\sum_{s=1}^p c_s W_s$ now follows from Lemma 3, as $p \rightarrow \infty$, which, by the consistency of E_i/p^2 , is also the required limit.

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